

BUI DUC SON

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EDUCATION

- Bachelor of Electronics and Telecommunications, Hanoi University of Science and Technology (GPA 3.48/4.00).
- Currently pursuing a combined Master's and Doctoral degree at Sungkyunkwan University (Rank 87th by Times Higher Education 2026).

RESEARCH DIRECTIONS

- Physical layer security of wireless communication systems.
- Adversarial attacks and defenses for advanced AI models.

PUBLICATIONS

- **Sep-2026:** H. Lee, **B. D. Son**, G. Zhao, B. G. Lee, D. I. Kim and I. Lee. "Behavior-Oriented Agent-Native Networks Toward B6G: From Delivering Information to Enabling Coordinated Decisions," preparing for IEEE Journal on Selected Areas in Communications.
- **Sep-2026:** G. Zhao, **B. D. Son**, J. Kim, and D. I. Kim. "Autonomous Digital Populations for Resilient Beyond-6G Edge Intelligence," preparing for IEEE Communications Magazine.
- **Aug-2026:** **B. D. Son**, T. V. Chien, D. I. Kim. "Adversarial Robustness of HGNN-aided Beamforming in Mobility-Aware STAR-RIS-Assisted MU-MISO Systems," submitted to IEEE Transactions on Communications.
- **Sep-2026:** M. Tran, T. N. Nguyen, **B. D. Son**, T. V. Chien, T. T. Duy, and M. Voznak "Unified Outage and Throughput Analysis for NOMA-MIMO Keyhole Channels under Imperfect SIC," minor revision, in IEEE Internet of Things Journal (Q1 - IF 8.9).
- **Aug-2026:** **B. D. Son**, T. V. Chien and D. I. Kim. "Adversarial Robustness of HGNN-Aided Beamforming in Dynamic STAR-RIS-Assisted MU-MISO Systems", IEEE Global Communications Conference 2026.
- **Nov-2025:** **B. D. Son**, G. Zhao, T. V. Chien and D. I. Kim. "Secure Distributed RIS-MIMO over Double Scattering Channels: Adversarial Attack, Defense, and SER Improvement", IEEE Transactions on Communications (Q1 - IF 8.3).
- **Aug-2025:** N. X. Tung, L. T. Giang, **B. D. Son**, S. G. Jeong, T. V. Chien, W. J. Hwang, and L. Hanzo. "Graph Neural Networks for Next-Generation-IoT: Recent Advances and Open Challenges.", in IEEE Communications Surveys & Tutorials (Q1 - IF 46.7).
- **Nov-2024:** **B. D. Son**, N. N. Khanh, T. V. Chien and D. I. Kim, "Adversarial Attacks Against Double RIS-Assisted MIMO Systems-Based Autoencoder in Finite-Scattering Environments," in IEEE Wireless Communications Letters (Q1 - IF 5.5).
- **Jun-2024:** **B. D. Son**, N. T. Hoa, T. V. Chien, W. Khalid, M. A. Ferrag, W. Choi, M. Debbah, "Adversarial Attacks and Defenses in 6G Network-Assisted IoT Systems," in IEEE Internet of Things Journal (Q1 - IF 8.9).
- **Jun-2023:** N. T. Hoa, L. V. Huy, **B. D. Son**, N. C. Luong and D. Niyato, "Dynamic Offloading for Edge Computing-Assisted Metaverse Systems," in IEEE Communications Letters (Q1 - IF 4.4).

ACHIEVEMENTS AND AWARDS

- **Feb-2026:** Distinguished Research Award, Fall 2025, CICE, Sungkyunkwan University.
- **May-2025:** Australia-Vietnam Strategic Technologies Scholarship for contribution in emergency technology.
- **Nov-2024:** Second prize in the Science and Technology Awards for Students in Universities 2024 (National level).
- **Jun-2024:** First prize in the 41st annual Scientific Research Student Conference for the 2023-2024 academic year in the field of Network Security - Digital Communications - Telecommunications, HUST (University level).

- **May-2024:** First potential award in the 41st annual Scientific Research Student Conference for the 2023-2024 academic year, SEEE, HUST (School level).
- **Oct-2023:** One of the top 10 outstanding interns after successfully passing two rigorous selection rounds in the “Viettel Digital Talent 2023” program in field 5G.

WORK EXPERIENCE

Feb 2025 - Now: Researcher - [WCL Lab, SKKU](#), under supervision of Prof. Dong In Kim

- Integrated Ph.D Course, Electrical and Computer Engineering. Research on the security of AI-aided 6G.
- I prepared the English version of the research proposal titled “Cross-Domain Adversarial Attack and Defense Mechanisms Development for AI-Native 6G Networks,” under the supervision of Prof. Dong In Kim, with assistance from my labmates. The proposal was successfully funded by the National Research Foundation (NRF) of Korea (RS-2026-25470140), \$450,000 March 2026 - February 2029.
- I prepared the English version of the research proposal titled “Next-Generation (B6G) Communication Systems through Evolution into Resilient and Autonomous Digital Populations,” under the supervision of Prof. Dong In Kim, with assistance from my labmates. The proposal was successfully funded by the National Research Foundation (NRF) of Korea (RS-2026-25608310), \$800,000 July 2026 - December 2030.
- Undergraduate intern supervisor: Supervising [Dong Joo Gu](#) on “Adversarial Attack against Autoencoder-Based Enhanced OTFS Modulation”.
- Undergraduate intern supervisor: Supervising [Tran Duc Toan](#) on “Revisiting Reconfigurable Intelligent Surface-aided Wireless Federated Learning: Adversarial Robustness Perspectives”, together with [Dr. Trinh Van Chien](#).

Jan 2025 - Now: Reviewer - IEEE JSAC, IEEE IoTJ, IEEE WCL, IEEE CL, IEEE PIMRC 2025, IEEE ICC 2026.

Sep 2022 - Feb 2025: Research assistant - [NETLAB, HUST](#) under supervision of Dr. Trinh Van Chien

- Research and write papers about Wireless Communication Systems that exploit AI.

Apr 2023 - Nov 2023: Viettel digital talent - Viettel High Technology

- Developed methods to detect Speedtest timing from end-user 5G device logs and created tools to measure 5G cell and UE quality, including BLER analysis from gNodeB logs to identify underperforming stations.

REFERENCE

- Assoc. Prof. Nguyen Tien Hoa, Lecturer, SEEE, Hanoi University of Science and Technology. [Google Scholar](#)
- Dr. Trinh Van Chien, Lecturer, SOICT, Hanoi University of Science and Technology. [Google Scholar](#)
- Prof. Dong In Kim, Director, Super Sapiens Research Institute, Sungkyunkwan University. [Google Scholar](#)

PROOF

SUNGKYUNKWAN UNIVERSITY
COLLEGE OF INFORMATION & COMMUNICATION ENGINEERING

DISTINGUISHED RESEARCH AWARD

BUI DUC SON

2025 FALL

PRESENTED ON THE FOURTH DAY OF FEBRUARY THE YEAR TWO THOUSAND AND TWENTY-SIX



DEAN

COLLEGE OF INFORMATION & COMMUNICATION ENGINEERING



우 수 상

전자전기컴퓨터공학과

석박사통합과정

BUI DUC SON

위 학생은 정보통신대학 우수연구심사에서 석사과정 우수상
수상자로 선정되었기에 이 상장과 부상을 수여함.

2026년 2월 4일



성균관대학교 정보통신대학장





Figure 2: Enter Caption



CỘNG HÒA XÃ HỘI CHỦ NGHĨA VIỆT NAM
Độc lập - Tự do - Hạnh phúc

**GIÁM ĐỐC ĐẠI HỌC BÁCH KHOA HÀ NỘI
TẶNG**

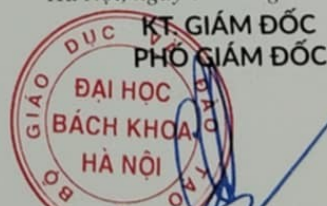
GIẤY KHEN

Sinh viên: **BÙI ĐỨC SƠN**
Mã số sinh viên: 20200524

**ĐẠT GIẢI NHÌ CẤP BỘ GIẢI THƯỞNG KHOA HỌC CÔNG NGHỆ
DÀNH CHO SINH VIÊN TRONG CƠ SỞ GIÁO DỤC ĐẠI HỌC
NĂM 2024**

**Tên đề tài: TẤN CÔNG VÀ PHÒNG THỦ TRONG MẠNG 6G-IOT SỬ DỤNG TRÍ TUỆ NHÂN
TẠO**

Hà Nội, ngày 04 tháng 6 năm 2025



Đã vào sổ khen thưởng
Số: 5496/QĐ-ĐHBK

PGS. TS. Huỳnh Đăng Chính

Figure 3: Enter Caption



Figure 4: Caption



Figure 5: Caption